

Executive Summary: Cell-Type-Specific Biomarkers in Human Liver Fibrosis

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1 Objective

Discover and prioritize **cell-type-specific biomarkers and therapeutic targets** in human liver fibrosis from single-cell transcriptomics (primary: GSE136103, cirrhotic vs healthy; validation: GSE244832, MASLD/MASH F0 to F4), focusing on the hepatic stellate/myofibroblast, macrophage/monocyte, and endothelial compartments.

2 Approach

A reproducible Snakemake + Scanpy pipeline ingests and QC's the data, integrates over donor (not disease) and verifies fibrosis signal is preserved, annotates and validates the three disease compartments, runs **donor-aware pseudobulk** differential expression, characterizes mechanism via pathway/TF activity and ligand-receptor analysis (with downstream-target corroboration), and prioritizes candidates with a transparent composite score combining DE effect, cell-type specificity, cross-dataset reproducibility, druggability, biomarker accessibility, and an explainable ML (XGBoost + SHAP) signal.

3 Top candidates

	rank	gene	compartment	composite	direction
0	1.0	PDGFRA	stellate_myofibroblast	0.671195	up
1	2.0	ACKR1	endothelial	0.634854	up
2	3.0	LUM	stellate_myofibroblast	0.624039	up
3	4.0	DCN	stellate_myofibroblast	0.611797	up

	rank	gene	compartment	composite	direction
4	5.0	TREM2	macrophage_monocyte	0.606202	up
5	6.0	PLVAP	endothelial	0.580863	up
7	7.0	COL3A1	stellate_myofibroblast	0.552956	up
9	8.0	MMP7	cholangiocyte	0.545108	up
10	9.0	SPP2	cholangiocyte	0.543718	up
11	10.0	MMP2	stellate_myofibroblast	0.542169	up
12	11.0	COL1A1	stellate_myofibroblast	0.537588	up
15	12.0	RGCC	endothelial	0.511566	up
16	13.0	COL1A2	stellate_myofibroblast	0.505817	up
17	14.0	RGS4	cholangiocyte	0.494848	up
22	15.0	VWF	endothelial	0.477451	up
33	16.0	FABP4	endothelial	0.431836	up
34	17.0	RGS10	macrophage_monocyte	0.428405	up
49	18.0	IER3	macrophage_monocyte	0.405677	up
78	19.0	PLXDC2	macrophage_monocyte	0.390093	up
118	20.0	CTSH	macrophage_monocyte	0.362371	up

4 Translational interpretation

The ranked table flags, per compartment, which candidates are most relevant for **diagnostics** (secreted/accessible, reproducible across datasets) versus **therapeutic targeting** (druggable, mechanistically central in the fibrotic crosstalk). See the full report and `results/08_score/candidate_scores.tsv` for the per-candidate rationale.

Full report and source code: <https://github.com/ericmalekos/sc-liver>